IN THE COMMAND CENTER on Earth, in a special room, you are helmeted and gloved. You turn your head to the left, and the cameras on the Mars robot rover turn to the left. You see, in very high definition and in color, what the cameras see. You take a step forward, and the rover walks forward. You reach out your arm to pick up something shiny in the soil, and the robot arm does likewise. The sands of Mars trickle through your fingers. The only difficulty with this remote reality technology is that all this must occur in tedious slow motion: The round-trip travel time of the up-link commands from Earth to Mars and the down-link data returned from Mars to Earth might take half an hour or more. But this is something we can learn to do. We can learn to contain our exploratory impatience if that's the price of exploring Mars. The rover can be made smart enough to deal with routine contingencies. Anything more challenging, and it makes a dead stop, puts itself into a safeguard mode, and radios for a very patient human controller to take over.

Conjure up roving, smart robots, each of them a small scientific laboratory, landing in the safe but dull places and wandering to view close-up some of that profusion of Martian wonders. Perhaps every day a robot would rove to its own horizon; each morning we would see close-up what had yesterday been only a distant eminence. The lengthening progress of a traverse route over the Martian landscape would appear on news programs and in schoolrooms. People would speculate on what will be found. Nightly newscasts from another planet, with their revelations of new terrains and new scientific findings, would make everyone on Earth a party to the adventure.

Then there's Martian virtual reality: The data sent back from Mars, stored in a modern computer, are fed into your helmet and gloves and boots. You are walking in an empty room on Earth, but to you you are on Mars: pink skies, fields of boulders, sand dunes stretching to the horizon where an immense volcano looms; you hear the sand crunching under your boots, you turn rocks over, dig a hole, sample the thin air, turn a corner, and come face to face with … whatever new discoveries we will make on Mars—all exact